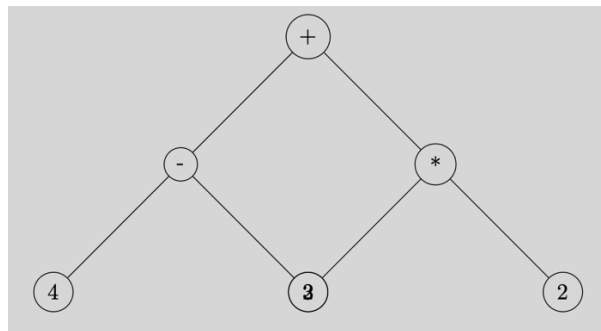
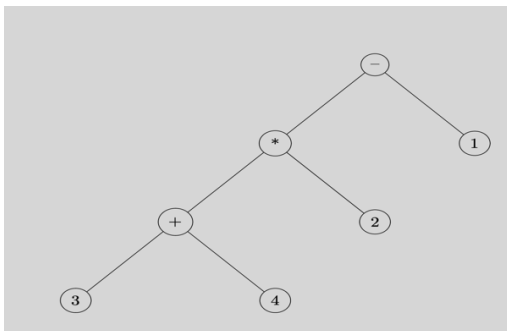


Evolutionary computing - Exercises

- Given two individuals: 010101 and 101010
 - Apply *2-point* crossover
 - Apply $(\mu+\lambda)$ selection with $f(x) = -x$
- Given two individuals: 11111111 and 00000000
 - Apply *3-point* crossover
 - Apply $(\mu+\lambda)$ selection with $f(x) = x$
- Consider two permutations A,B,C,D,E,F and F,E,D,C,B,A
 - Apply *order 1* crossover on a subset of three elements
 - To one of the children apply *swap* mutation and to the other one *insert* mutation
- Repeat the previous exercise on the two permutations: 1,3,5,2,4,6 and 2,4,6,1,3,5.
- Consider the two trees below, representing, respectively, the two formulae $((3+4)*2)-1$ and $(4-3)+(2*2)$



Apply crossover. Which one is the most fit individual if $f(T) = |\text{evaluate}(T) - 10|$? [evaluate(T) is the result of evaluating the formula represented by the tree].

- Consider the two logical expressions $(x \wedge y) \vee (x \rightarrow y)$ and $(y \vee x) \rightarrow x$
 - Represent them by means of a binary tree
 - Compute their fitness with $f(T)$ = number of truth assignments for which the expression represented by T evaluates to true
 - Apply crossover
- Let us consider an individual represented by 4 binary digit, for instance 1011, labeled from right to left as $x_0 = 1, x_1 = 1, x_2 = 0, x_3 = 1$. The fitness function is defined as $f(x) = x_3 + 2x_2 + 6x_1 + x_0$. Apply simple GA evolution:
 - Generate random initial population of 4 individuals
 - Generate 4 new individuals: parents selected by roulette wheel selection, 1-point crossover and bit-wise mutation with $p_m = 1/4$
 - Survivor selection: age based
 - Create a new offspring with tournament selection and $k=2$

- Given two individuals: 010101 and 101010

- Apply 2-point crossover
- Apply $(\mu + \lambda)$ selection with $f(x) = -x$

IND.1 010101
IND.2 101010 $\Rightarrow$ IND.3 011001
IND.4 100101

$$\begin{aligned} f(\text{IND.1}) &= 2^0 + 2^2 + 2^4 = 21 \Rightarrow f(\text{IND.1}) = -21 \\ f(\text{IND.2}) &= 2^1 + 2^3 + 2^5 = 12 \Rightarrow f(\text{IND.2}) = -12 \\ f(\text{IND.3}) &= 2^0 + 2^3 + 2^4 = 25 \Rightarrow f(\text{IND.3}) = -25 \\ f(\text{IND.4}) &= 2^0 + 2^2 + 2^5 = 38 \Rightarrow f(\text{IND.4}) = -38 \end{aligned}$$

IND.1 AND IND.3 BEST INDIVIDUAL

Algorithm 5.1: Non-Self-adaptive Evolution Strategies $(\mu + \lambda)$

Randomly create an initial population $\{x^1, \dots, x^\mu\}$ of parent vectors, where each x^i is of the form $x^i = (x_1^i, \dots, x_n^i)$, $i = 1, \dots, \mu$;
Evaluate each member of the population;
repeat
 Randomly select (individual) parent from $\{x^i : i = 1, \dots, \mu\}$;
 Create child vector by applying a mutation operator to the parent;
until λ children are generated;
Rank the $\mu + \lambda$ parents and children from best (those producing least error on the training dataset) to worst;
Select the best μ of these to continue into the next generation;
until terminating condition;

- Given two individuals: 11111111 and 00000000
 - Apply 3-point crossover
 - Apply $(\mu + \lambda)$ selection with $f(x) = x$
- Consider two permutations A,B,C,D,E,F and F,E,D,C,B,A

IND.1 11111111 $f(\text{IND.1}) = (2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7) = 255$
IND.2 00000000 $f(\text{IND.2}) = 0$
IND.3 10001110 $f(\text{IND.3}) = (2^7 + 2^3 + 2^2 + 2^1) = 142$
IND.4 01110001 $f(\text{IND.4}) = (2^6 + 2^5 + 2^4 + 2^0) = 113$

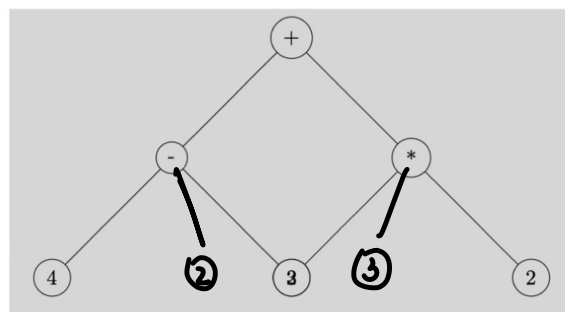
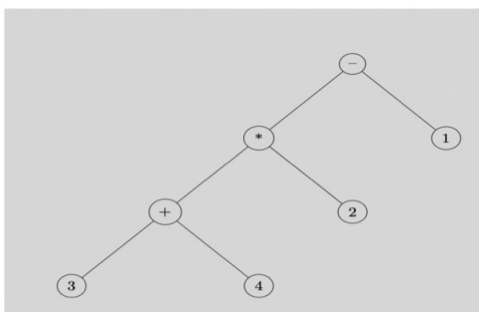
CHOOSE IND.1 AND IND.3

- Consider two permutations A,B,C,D,E,F and F,E,D,C,B,A
 - Apply order 1 crossover on a subset of three elements
 - To one of the children apply swap mutation and to the other one insert mutation
- Repeat the previous exercise on the two permutations: 1,3,5,2,4,6 and 2,4,6,1,3,5.

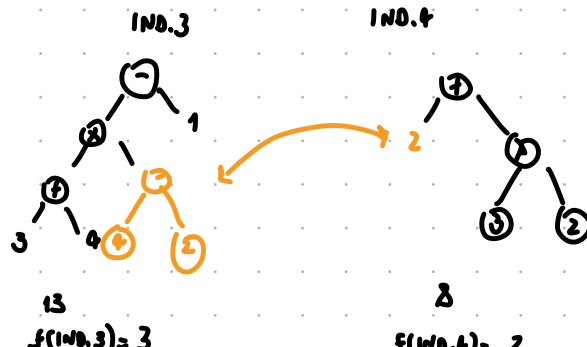
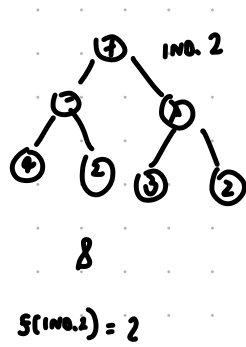
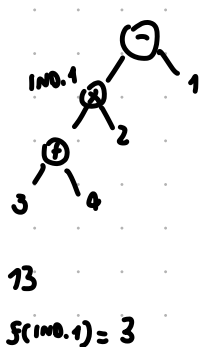
ABCDEF $\xrightarrow{\text{SWAP}}$ EBCDAF $\xrightarrow{\text{SWAP}}$ FBCEAD
FEDCBA $\xrightarrow{\text{INSERT}}$ BEFCFA $\xrightarrow{\text{INSERT}}$ BDECFB

135246 $\xrightarrow{\text{SWAP}}$ 135246 $\xrightarrow{\text{SWAP}}$ 132546
246135 $\xrightarrow{\text{INSERT}}$ 246135 $\xrightarrow{\text{INSERT}}$ 243615

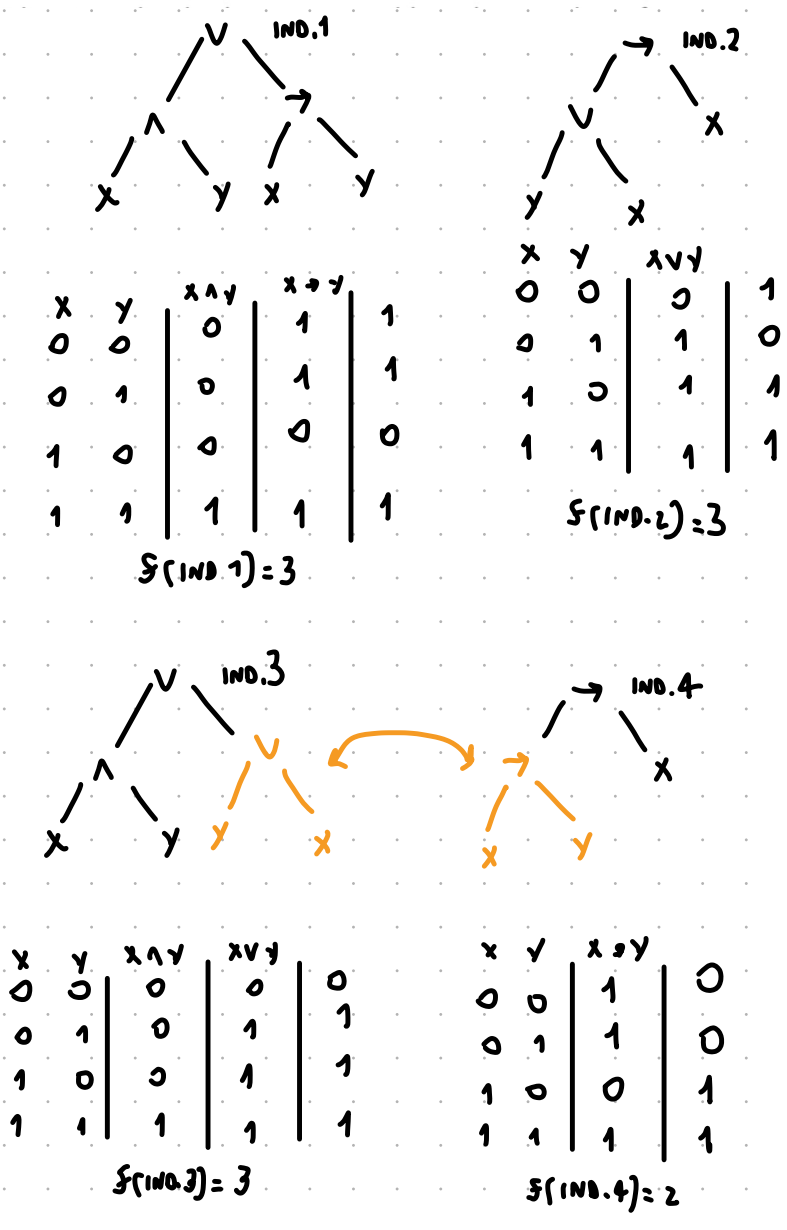
- Consider the two trees below, representing, respectively, the two formulae $((3+4)*2)-1$ and $(4-3)+(2*2)$



Apply crossover. Which one is the most fit individual if $f(T) = |\text{evaluate}(T) - 10|$? [evaluate(T) is the result of evaluating the formula represented by the tree].



- Consider the two logical expressions $(x \wedge y) \vee (x \rightarrow y)$ and $(y \vee x) \rightarrow x$
 - Represent them by means of a binary tree
 - Compute their fitness with $f(T)$ = number of truth assignments for which the expression represented by T evaluates to true
 - Apply crossover



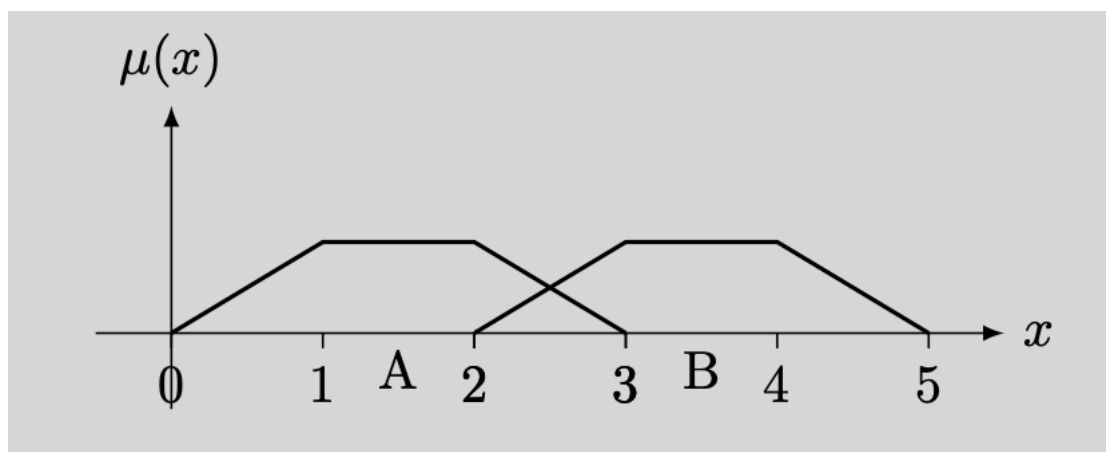
Fuzzy Systems and Evolutionary Computing 2022/23

Open Questions

- What is a linguistic variable? Give an example.
- What are fuzzification and defuzzification in a fuzzy rule based system? Which is the connection with a frame of cognition (aka, fuzzy partition)?
- Explain how fuzzy c-means works. If helpful, you can consider the two centroids $c_1 = (0, 1)$, $c_2 = (1, 0)$, $w = 2$ and three objects $x_1 = (0.75, 0.5)$, $x_2 = (0, 0)$ and $x_3 = (0.3, 1.1)$.
- Define n-point crossover. Give an example.
- Mutation in EA: what is it? Which is its role? Give an example with integer representation
- Give a high-level (=no need of mathematical details) explanation of Particle Swarm Optimization

Exercises

- 1) Let us consider the two fuzzy sets A and B given in the following figure



- a) Give the equation of the two functions defining A and B
- b) Compute the intersection and union of f and g by means of a t-norm/t-conorm at your choice
- c) Compute the complement of A
- d) Compute support and core of G
- 2) Let us consider the fuzzy set g defined as $g(5)=0.3$, $g(6)=0.5$, $g(7)=0.9$, $g(8)=1$, $g(9)=0.9$, $g(10)=0.5$, $g(11)=0.3$ $g(x) = 0$ for all other values of x.
- a) Compute core and support of g
- b) Compute the complement of g
- c) Compute the cardinality of g
- d) Compute the α -cut of g with $\alpha=0.6$
- e) Compute the entropy of g
- f) Compute the measure of fuzziness of g using the Euclidean distance
- g) Compute the ordered weighted average with $w_1 = w_2 = w_3 = 2/14$, $w_4 = w_5 = 1/14$, $w_6 = w_7 = 3/14$
- h) Compute the center of gravity of g

- 3) Given two propositions P and Q, with truth value $v(P)=0.7$ and $v(Q)=0.4$, compute
 - a) The logical *and* using product t-norm
 - b) The logical *or* using product t-conorm
 - c) $P \rightarrow Q$, using an implication at your choice
- 4) Let us consider a population of integer expressions (for instance $(3+4)-(6+8)$) with operations in $\{+, -, *, /\}$ and numbers in $\{0, \dots, 9\}$. Apply simple Genetic Programming with the following parameters
 - a) Random initial population of 4 individuals generated with full method and depth = 3
 - b) The fitness function is defined as $f(x) = x + 2$ with x the result of the expression represented by the tree
 - c) Generate 4 new individuals: two are obtained by crossover of the two most fit parents and two by mutation of the other two parents
 - d) Survivor selection: generational replacement
- 5) Let us consider a population of individuals represented by 6 binary digits. Apply simple Genetic Programming with the following parameters:
 - a) Generate random initial population of 4 individuals
 - b) Generate 4 new individuals: parents selected by roulette wheel selection, 2-point crossover and bit-wise mutation with $p_m = 1/6$
 - c) Survivor selection: age based
 - d) Create a new offspring with tournament selection and $k=2$

